General Biology 1

First Semester

Supplementary Materials

Module 2

The Cell

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What I Know

Directions: Read each situation carefully and choose the best answer.

1. Which organelle is considered as the power house of the cell?
   a. Cytoplasm
   b. Mitochondrion
   c. Lysosome
   d. Cell Membrane
2. Which organelle is the brain of the cell?
   a. Ribosomes
   b. Vacuole
   c. Nucleus
   d. Endoplasmic Reticulum
3. Which organelle contains digestive enzymes that destroy foreign materials in the cell?
   a. Lysosome
   b. Mitochondrion
   c. Nucleus
   d. Ribosome
4. Which organelle is responsible for the synthesis of phospholipids and steroids?
   a. Nucleus
   b. Ribosome
   c. Rough endoplasmic reticulum
   d. Smooth endoplasmic reticulum
5. Which of the following refers to the function of chloroplasts?
   a. Lipid Synthesis
   b. Photosynthesis
   c. Protein Synthesis
   d. Cellular Respiration

What I Need to Know

This supplementary material is designed to support and enhance students' understanding of the core concepts in biology. These materials provide additional explanations, activities, examples, and assessments that help reinforce the lessons discussed in class or presented in the main learning module.

Lesson 2 The Cell

In the previous lesson, you learned about the origin of cells and the three postulates of the Cell Theory. In this lesson you will learn the structure and function of each organelle and how they carry out essential functions needed for growth, development, and survival. Cells contain small membrane-bound structures called **organelles** that performs specific functions.

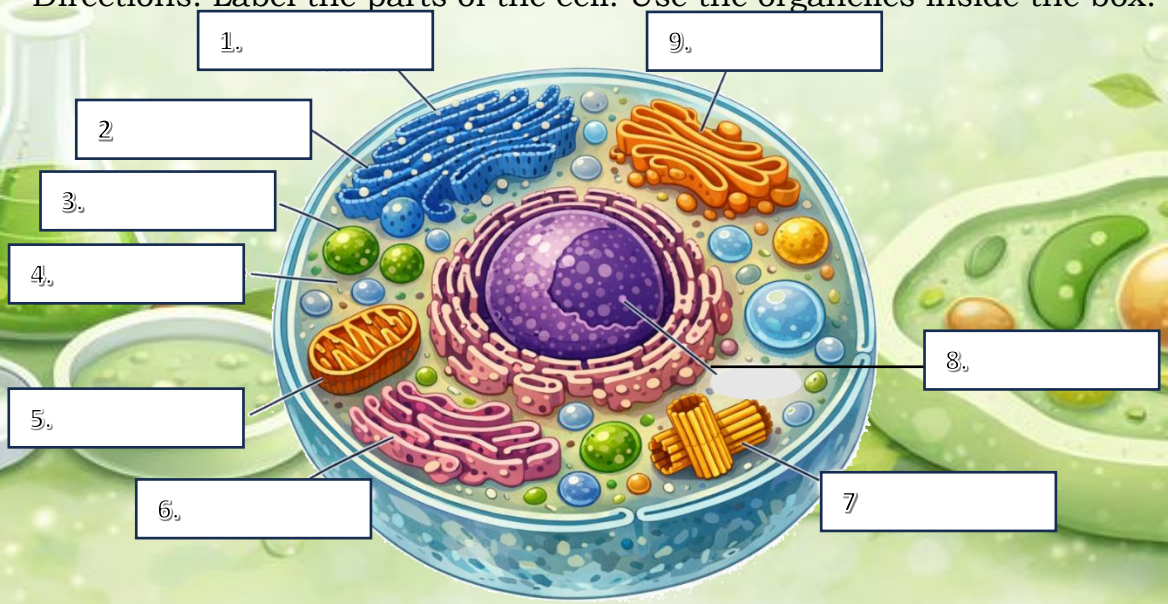
After going through this supplementary module, you are expected to:

1. identify the different cell organelles;
2. describe the structure and functions of the cell organelles; and
3. relate the function of cell organelles in city-like illustration.

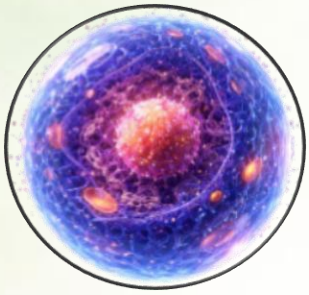
What's In

Activity 1: "Cell Explorer: Identify the Organelles"

Directions: Label the parts of the cell. Use the organelles inside the box.



Vesicle	Centrioles	Lysosome
Nucleus	Rough ER	Peroxisomes
Golgi Apparatus	Smooth ER	Mitochondrion



Nucleus

The nucleus is important to the life of the cell. It is commonly known as the "brain of the cell". It contains and stores all the genetic information for manufacturing proteins and responsible for controlling all cellular activities.

- The nucleus consists of chromatin, nucleolus, and nuclear envelope.
- The chromatin is a network of strands that condenses and undergoes coiling into rod-like structures called chromosomes. Chromatin contains DNA (Deoxyribonucleic Acid), histones and some RNA (Ribonucleic Acid).
- The nucleolus is a dark region of chromatin where a type of RNA, called ribosomal RNA (rRNA), is produced, and joins with proteins to form the subunits of ribosomes.
- The nuclear envelope is a double membrane that separates the nucleus from the cytoplasm. It has nuclear pores to permit the passage of RNA out of the nucleus into the cytoplasm, and the passage of proteins from then cytoplasm into the nucleus.
- The nucleus serves as the storage of genetic materials such as DNA and RNA.
- Nucleus controls other organelles by facilitating cell growth, cell division, and protein synthesis.

Ribosomes

- The ribosomes are also known as the "protein factories of the cell".
- They are mainly responsible for protein synthesis in the cells.
- They are usually dispersed in the cytoplasm or attached to the endoplasmic reticulum.
- Ribosomes are composed of one large subunit and one small subunit.

- The number of ribosomes in a cell varies depending on its functions. For example, pancreatic cells have many ribosomes because they produce secretions that contain proteins.
- Ribosomes receive mRNA from the nucleus, and this nucleic acid carries a coded message from DNA. The mRNA carries the correct sequence of amino acids in a protein. Protein synthesized by the cytoplasmic ribosomes are used in the cytoplasm, while proteins synthesized by attached ribosomes end up in the endoplasmic reticulum.

The Endomembrane System

The endomembrane system consists of the three major organelles that work together to perform different tasks such as packaging, producing, and exporting certain cellular products. The endomembrane system includes the endoplasmic reticulum, Golgi apparatus and several types of vesicles.

- The endoplasmic reticulum (ER) consists of complicated system of membranous channels and flattened vesicles which is physically continuous to the outer membrane of the nuclear envelope.
- There are two types of endoplasmic reticulum: the rough endoplasmic reticulum and smooth endoplasmic reticulum.
- Rough endoplasmic reticulum is studded with ribosomes on the side of the membrane that faces the cytoplasm. It is responsible for synthesizing proteins.
- Rough endoplasmic reticulum forms vesicles in which large molecules are transported to other parts of the cell.
- Smooth endoplasmic reticulum which is continuous with rough ER, does not have attached ribosomes. Smooth endoplasmic reticulum is responsible for synthesizing lipids, such as phospholipids and steroids. The function of smooth ER is usually dependent on a particular cell. For example, smooth ER in the liver helps in detoxifying drugs. Usually, smooth ER increases its quantity when a person regularly consumes alcohols.

Golgi Apparatus (Dictyosomes)

- The Golgi apparatus are a stack of curved, flattened sacs, which can be compared to a pile of pancakes.
- Mostly, one side of the stack (the cis) is directed towards the endoplasmic reticulum, and the other side of the stack (the trans) is directed towards the cell membrane.
- The Golgi apparatus receives vesicles from the endoplasmic reticulum along the cis face. Once inside the Golgi apparatus, the material is altered in the lumen of the vesicle. The vesicles then start to swell and be released from the trans face. These vesicles may be either exported outside the cell or may be used by the cell for its own use.

Lysosomes

- The lysosomes are membrane-bounded vesicles produced by the Golgi apparatus.
- Lysosomes are very acidic and contain powerful hydrolytic digestive enzymes that require a low–pH environment. Hydrolytic enzymes break down large molecules into small molecules. For example, they break down large carbohydrates into simple sugars that will serve as a nutrient for the cell.
- Lysosomes are responsible in recycling cellular material and destroying invading viruses and bacteria.
- Lysosomes also participate in a programmed cell death called apoptosis which is a normal part of development. For example, the fingers of human embryo are at webbed at first, but eventually freed from one another because of lysosomal action.

Energy–Related Organelles

One of the functions of the cell is to convert energy to a form that can be used by the cell. It includes organelles such as mitochondria and chloroplasts.

Mitochondrion

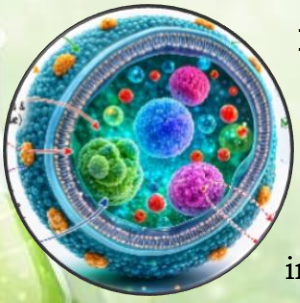
- The mitochondrion (mitochondria, plural) is often called the “powerhouse of the cell.”
- It produces most of the ATP (Adenosine Triphosphate) which is used by the cell through the process called cellular respiration.

- The mitochondrion is a double membrane-bound organelle that is responsible for supplying and storing energy for the cell. It has two membranes, the outer membrane and the inner membrane. The inner membrane is highly convoluted into cristae that project into the matrix. The inner membrane surrounds a matrix containing its own mitochondrial DNA and ribosomes. The matrix is a highly concentrated mixture of enzymes that break down carbohydrates and other nutrient molecules.
- The mitochondrion is where the synthesis of energy in the form of ATP takes place which is required for other cell organelles to function.

- x Mitochondria also help in the balancing of cations within the cell and assists the process of apoptosis. Apoptosis is also known as “cellular suicide”. It is a convenient way of removing cells that should not be longer part of the cell.
- x Mitochondria in the liver can detoxify ammonia.

Chloroplasts

- Chloroplasts are plant and algal organelles that are bounded by a double membrane.
- A chloroplast contains a series of internal membrane or vesicles. It contains DNA and chlorophyll which carry on photosynthesis. Its double membrane encloses a large space called stroma.



Peroxisomes

- The peroxisomes are membrane-bounded vesicles that encloses enzymes.
- The enzymes in the peroxisomes are synthesized by free ribosomes and transported into the cytoplasm.
- Each peroxisome also carries enzymes which results in the formation of hydrogen peroxide (H2O2).
- Peroxisomes can also detoxify some products of photosynthesis.

What's New

Activity 2: "Cell City Analogy"

Directions: Look at the city-like illustration of the cell. Identify the correct cell organelle corresponding to the city roles.

Cell Analogy			
	City Analogy	**Cell Organelle**	**Function/City Roles**
1	City Hall		
2	Power Plant		
3	Factory		
4	Storage Warehouse		

5	Security Gate		
6	Post Office		
7	Transport System		

Guide Questions:

1. If the mitochondria stopped working, what would happen to the cell? How is this similar to a city losing its power supply?

   _____

   _____

2. What might happen to a city if the government (like the nucleus) stopped functioning properly? How would this affect the cell?

   _____

   _____

3. Why is it important that all organelles work together to keep the cell alive?

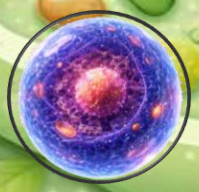
   _____

   _____

What's More

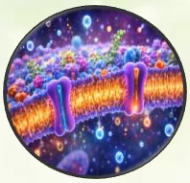
Activity 3: "Cell Detective Activity"

Directions: Be a DETECTIVE, read each clue carefully to solve this case by cutting each organelle and place it accordingly to the magnifying lens.

CLUE	ORGANELLE
1. I am the control center of the cell. I contain genetic material that directs all activities.	
2. I produce energy that powers the cell's activities.	
3. I control what enters and leaves the cell.	

4. I package and distribute proteins to different parts of the cell.

5. I am the site where proteins are produced.



To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A toxin damages the mitochondria of a cell. Which cellular process will be most directly affected?
   a. Protein synthesis b. Energy production
   c. Lipid storage d. DNA replication
2. Which cell organelle is responsible for protein synthesis?
   a. Mitochondria b. Chloroplasts
   c. Nucleus d. Ribosomes
3. A scientist damages the Golgi apparatus in a cell. Which function will most likely be affected?
   a. Modification and packaging of proteins b. Production of ATP
   c. Photosynthesis d. DNA storage
4. Which cell organelle is very acidic and contains powerful hydrolytic digestive enzymes that require a low–pH environment?
   a. Nucleus c. Endoplasmic
   c. Lysosomes d. Ribosomes
5. In a city-like structure of a cell, the cell membrane serves as a security gate. Which of the following is the correct explanation of this analogy?
   a. Because the cell membrane produces energy for the cell
   b. Because the cell membrane produces proteins
   c. Because the cell membrane controls what enters and leaves the city just like the cell membrane regulates materials entering and exiting the cell.
   d. Because the cell membrane breaks down waste materials and old structures in the city